

Requirements

Python Version

Python **3.12.10** (tested). Python 3.9 or higher should be compatible.

Dependencies

Exact versions used to produce the reported results:

Package	Tested version	Minimum	Purpose
tensorflow	2.21.0	≥ 2.10	Loading and querying pre-trained DNN models (.h5)
keras	3.14.0	≥ 3.0	Bundled with TensorFlow; model loading via <code>keras.models.load_model</code>
ortools	9.15.6755	≥ 9.0	CP-SAT constraint solver (CSP and Hybrid Phase 2)
scikit-learn	1.8.0	≥ 1.0	DecisionTreeClassifier (DT and Hybrid Phase 1)
pandas	3.0.2	≥ 1.5	Dataset loading and manipulation
numpy	2.4.4	≥ 1.23	Numerical operations
scipy	1.17.1	≥ 1.9	Wilcoxon signed-rank test (<code>scipy.stats.wilcoxon</code>)
matplotlib	3.10.8	≥ 3.6	Optional — results visualisation
pytest	9.0.3	≥ 7.0	Test suite (135 tests)

Installation

```
pip install -r requirements.txt
```

All packages are available on PyPI. No conda or special channels required.

Hardware

No GPU required. All experiments run on CPU only. Approximate wall-clock runtime per dataset for a 20-run DT experiment (budget=1000, standard laptop, Intel CPU):

Dataset	Baseline (20 runs)	DT (20 runs)	Total (both)
adult	~44 min	~43 min	~87 min
compas	~42 min	~42 min	~84 min

Dataset	Baseline (20 runs)	DT (20 runs)	Total (both)
dutch	~39 min	~39 min	~78 min
credit	~39 min	~39 min	~78 min
german	~47 min	~40 min	~87 min
kdd	~40 min	~44 min	~84 min

CSP and Hybrid methods are slower due to CP-SAT solver overhead (~197s/run observed on German).

Running `--dataset all --method all` covers all 8 datasets (including Law School and Communities & Crime ablation datasets) and will take several hours — best left as an overnight run. To run only the 6 primary datasets, specify them individually or as a comma-separated list (e.g. `--dataset kdd,adult,compas,dutch,credit,german`).
